

Software Requirements Specification

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****Software Requirements Specification****

****1. Introduction****

This Software Requirements Specification (SRS) document outlines the requirements for the B09 application, a system designed to manage and track user activities, badges, and leaderboards. The application consists of a backend API, a frontend web interface, and a mobile application.

****2. Purpose****

The purpose of this SRS is to define the functional and non-functional requirements of the B09 application, ensuring that all stakeholders have a clear understanding of the system's capabilities and limitations.

****3. Scope****

The scope of this project includes the development of the B09 application, comprising the following components:

- * Backend API: responsible for managing user data, activities, badges, and leaderboards.
- * Frontend web interface: provides a user-friendly interface for users to interact with the application.
- * Mobile application: allows users to access the application on their mobile devices.

****4. System Overview****

The B09 application is designed to manage and track user activities, badges, and leaderboards. The system consists of the following components:

- * User Management: allows users to register and manage their accounts.
- * Activity Management: enables users to complete activities and earn badges.
- * Badge Management: manages the issuance and tracking of badges.
- * Leaderboard Management: displays user rankings and leaderboards.

****5. Functional Requirements****

The B09 application shall meet the following functional requirements:

- * User Registration: The system shall allow users to register and create an account.
- * Activity Completion: The system shall enable users to complete activities and earn badges.
- * Badge Issuance: The system shall issue badges to users upon completing activities.
- * Leaderboard Display: The system shall display user rankings and leaderboards.
- * API Calls: The system shall handle API calls for activities, user registration, and leaderboard data.

****6. Non-Functional Requirements****

The B09 application shall meet the following non-functional requirements:

- * Performance: The system shall respond to user requests within 2 seconds.
- * Security: The system shall ensure user data is encrypted and protected against unauthorized access.
- * Usability: The system shall provide an intuitive and user-friendly interface.
- * Scalability: The system shall be able to handle a minimum of 1000 concurrent users.

****7. Architecture****

The B09 application shall consist of the following architecture:

- * Backend API: built using Node.js, Express, and MongoDB.
- * Frontend web interface: built using React and JavaScript.
- * Mobile application: built using React Native and JavaScript.
- * Database: MongoDB shall be used to store user data, activities, badges, and leaderboards.

****8. Risks****

The following risks have been identified:

- * Data security breaches
- * System downtime
- * Compatibility issues with different devices and browsers
- * Scalability issues with increasing user traffic

****9. Constraints****

The following constraints have been identified:

- * The system shall be deployed on a cloud-based platform (Vercel).
- * The system shall use MongoDB as the database management system.
- * The system shall be developed using Agile methodology.

****10. Future Scope****

Future developments may include:

- * Integration with social media platforms

- * Addition of new features, such as gamification and challenges
- * Expansion of the system to support multiple languages and regions
- * Development of a desktop application

Recent developments have included updates to the vercel.json file, addition of new routes and API calls, and deployment of the backend and mobile applications. The system has also been updated to include DailyQuest automatic task completion and index redirects.

Repository Context:

The repository contains the following files:

- * backend/models/Activity.js: Mongo model for activities
- * backend/server.js: API calls for activities
- * backend/routes/userRoutes.js: Express routes for user registration and activity completion
- * backend/models/Badge.js: Mongo model for badges
- * backend/routes/activityRoutes.js: Express routes for activity management
- * mobile/app/(tabs)/Activities.tsx: React component for activities screen
- * frontend/src/pages/ActivitiesPage.jsx: API call for activities
- * backend/routes/leaderboardRoutes.js: Express route for leaderboard display

These files provide the foundation for the B09 application and shall be used as a reference for future developments.